

Coordinate systems

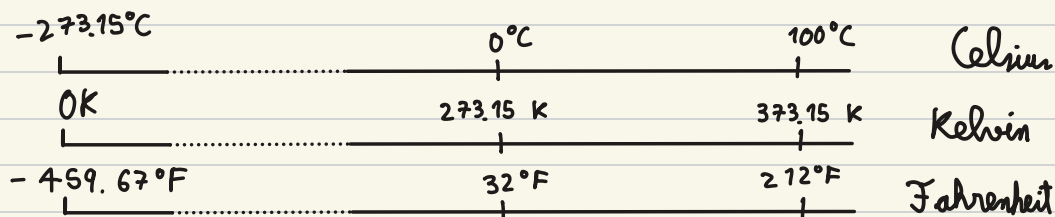
April 13, 2026

A coordinate system is like the scale and reference points in a map, it tells us how some numbers correspond to a location (in a line, a curve, a surface, etc.)

But we can also use coordinates to identify numbers with other quantities:
eg: Temperature, brightness (light intensity), Forces, ...

We can use many different coordinate systems to describe the same object:

eg: Temperature



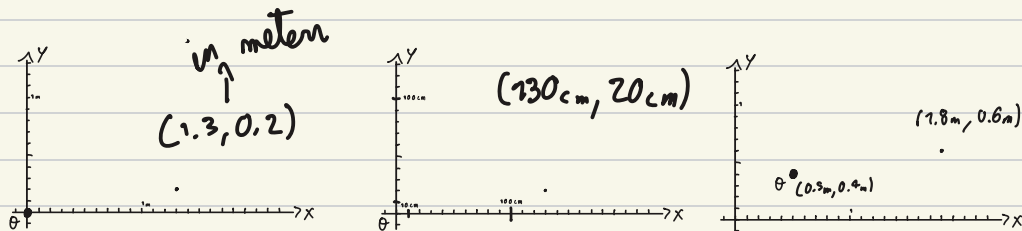
T in Celsius $\rightarrow$ T in Kelvin
C K

$$C + 273.15 = K$$

T in Celsius $\rightarrow$ T in Fahrenheit $F = 1.8 \cdot C + 32$

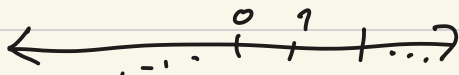
These are examples of affine or linear change of coord

e.g. Describing location from an origin point to some point

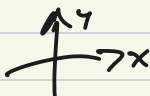


Our basic coordinate system

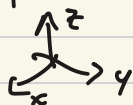
Cartesian: in 1-D is just the real line



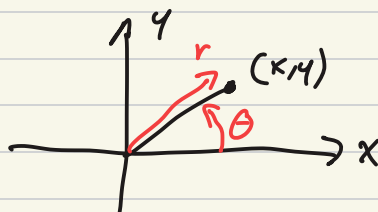
2-D



3-D



Polar coordinates: in 2-D

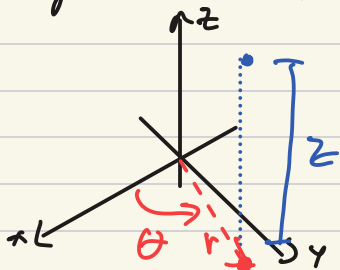


$$x = r \cos \theta$$

$$y = r \sin \theta$$

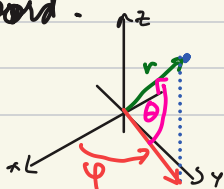
$$(r, \theta) \text{ in polar coord} \longleftrightarrow (r \cos \theta, r \sin \theta) \text{ in cartesian coord}$$

in 3-D: cylindrical coord.



$$(r, \theta, z) \text{ in cylindrical} \longleftrightarrow (r \cos \theta, r \sin \theta, z) \text{ in cartesian}$$

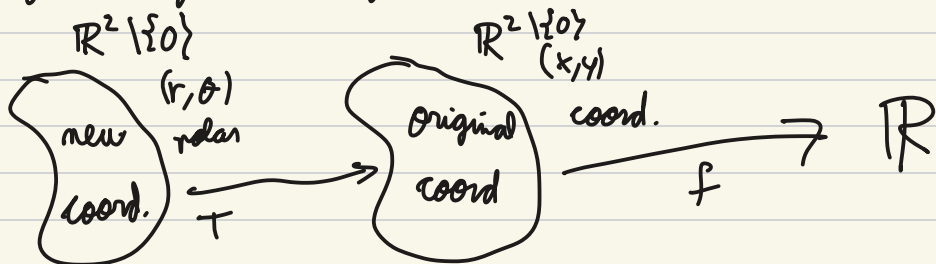
in 3-D: polar coord.



$$(r, \theta, \varphi) \text{ in polar} \longleftrightarrow \begin{pmatrix} r \cos \theta \cos \varphi \\ r \cos \theta \sin \varphi \\ r \sin \theta \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ in cartesian}$$

How does change of coordinates "affect calculus"

• Rate of change : By the chain rule



eg: $f(x, y) = x^2 + y^2$

$T(r, \theta) = (r \cos \theta, r \sin \theta)$
change of coord.

$$\vec{\nabla}(f \circ T)|_{(r, \theta)} = D(f \circ T) = Df|_{T(r, \theta)} \cdot DT|_{(r, \theta)}$$

$$\begin{aligned} f \circ T(r, \theta) &= f(r \cos \theta, r \sin \theta) \\ &= r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2 \end{aligned}$$

$$\vec{\nabla}(f \circ T)|_{(r, \theta)} = (2r, 0)$$

$$\vec{\nabla}f = (2x, 2y) \quad DT = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix}$$

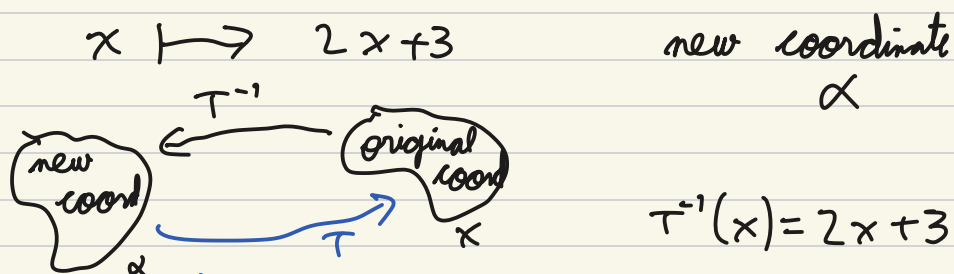
"

$$\begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

$$\begin{aligned} (2r \cos \theta, 2r \sin \theta) \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} &= (2r \cos^2 \theta + 2r \sin^2 \theta, -2r \cos \theta \sin \theta + 2r \sin \theta \cos \theta) \\ &= (2r, 0) \end{aligned}$$

• Change of length, area and volume

- Affine transformation in $\mathbb{R}$



length of interval $[0, 2]$ in x is 2
in new coord. $[0, 2]$ becomes $[3, 7] = [c, d]$

the length of the interval in terms of the new coordinate is

$$\left(\frac{1}{2}\right) \cdot (d - c) = \frac{7-3}{2} = 2$$

is coming from $\frac{dT}{d\alpha}$

$$T(\alpha) = \frac{\alpha-3}{2} \quad \frac{dT}{d\alpha} = \frac{1}{2}$$

each unit of distance in the new coordinate is $\frac{1}{2}$ unit of distance in original coord

coordinate

* Addendum: An affine transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$, is a transformation of the form "scaling" and translating. This means

$$T(\vec{x}) = A\vec{x} + \vec{b} \quad \text{where } \vec{b} \in \mathbb{R}^n, A \in M_{n \times n}(\mathbb{R}) \text{ and } \det(A) \neq 0.$$

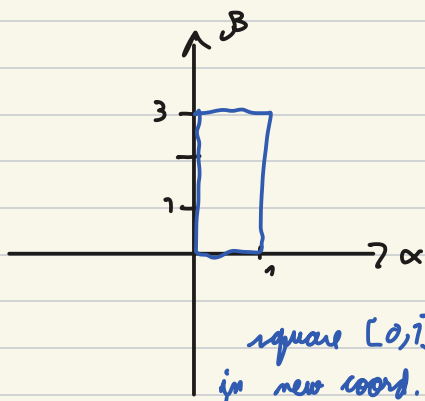
in 2-D: $\alpha = 2x + 3$, $\beta = x + 3y$

orig. coord. = (x, y) new coord. = (α, β)

$$T^{-1}(x, y) = (\alpha, \beta) = (2x + 3, x + 3y)$$

$$T(\alpha, \beta) = (x, y) = \left(\frac{\alpha - 3}{2}, \left(\beta - \frac{\alpha - 3}{2} \right) / 3 \right)$$

$\times$



(α, β)
 $(0, 0)$
 $(1, 0)$
 $(1, 3)$
 $(0, 3)$

(x, y)
 $\left(-\frac{3}{2}, \frac{1}{2}\right)$
 $\left(-1, \frac{1}{3}\right)$
 $\left(-1, \frac{4}{3}\right)$
 $\left(-\frac{3}{2}, \frac{3}{2}\right)$

Area in (α, β) is 3

Area in (x, y) is $\det \begin{pmatrix} -1 + \frac{3}{2} & \frac{1}{3} - \frac{1}{2} \\ -\frac{3}{2} - \frac{3}{2} & \frac{3}{2} - \frac{1}{2} \end{pmatrix}$

$$= \det \begin{pmatrix} \frac{1}{2} & -\frac{1}{6} \\ 0 & 1 \end{pmatrix} = \frac{1}{2}$$

$$DT = \begin{pmatrix} \frac{\partial x}{\partial \alpha} & \frac{\partial x}{\partial \beta} \\ \frac{\partial y}{\partial \alpha} & \frac{\partial y}{\partial \beta} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{1}{6} & \frac{1}{3} \end{pmatrix} = \frac{1}{6}$$